

REPOSITORY GUIDE

XAI-Driven Modality Bias Detection and Mitigation

Concise Hinglish Technical Guide

Core idea: multimodal model me ek easy modality dominate kar sakti hai. Repository bias ko attribution aur discrepancy ratio se measure karti hai, phir OPM aur OGM-GE se dominant branch ko training ke dauran regulate karti hai.

IMPLEMENTED FOCUS

DETECT	QUANTIFY	MITIGATE	VERIFY
Attention + XAI	Discrepancy rho	OPM + OGM-GE	Metrics + plots

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Contents

Guide ko first-to-last read kiya ja sakta hai, ya specific section directly use kiya ja sakta hai.

1. One-minute summary	9. Main results
2. Problem intuition	10. XAI and verification tools
3. High-level architecture	11. How to run the project
4. Core measurement: discrepancy ratio	12. Output files
5. Mitigation methods	13. What is implemented vs proposed
6. Exact training flow	14. Limitations
7. Repository map	15. Short explanation script
8. Datasets and experimental setup	16. Key terms
	Reference

Fast reading path

Sections 1 → 3 → 5 → 8 → 9 → 13 → 15 cover the complete project in the shortest path.

1. One-minute summary

This project studies **modality bias** in multimodal models. A multimodal model may receive two useful inputs—for example, an image and text—but learn to depend mainly on the easier modality. The other encoder then remains under-trained, making the model less balanced and potentially less robust.

The repository has two connected parts:

- 1 **Detection / explanation:** PaliGemma attention visualization, image Integrated Gradients, text gradient $\times$ input, uni-modal accuracy, and modality attribution shares.
- 2 **Mitigation:** OPM and OGM-GE dynamically slow down the dominant modality during training so that the weaker encoder receives a better learning opportunity.

The main implemented pipeline is in [balanced_mm/](#). It supports image + text, audio + text, synthetic data, Food-101, and MELD. Experiments compare four settings: no modulation, OPM, OGM-GE, and both methods together.

Main finding: modulation clearly reduces modality discrepancy. It also improves accuracy on the synthetic and Food-101 settings, but on MELD it reduces imbalance while slightly reducing accuracy because text is much stronger than the available audio features.

2. Problem intuition

Assume the model receives two modalities:

- image feature: ϕ_{image}
- text feature: ϕ_{text}

A late-fusion classifier combines both:

$$\text{logits} = W_{\text{image}} * \phi_{\text{image}} + W_{\text{text}} * \phi_{\text{text}} + \text{bias}$$

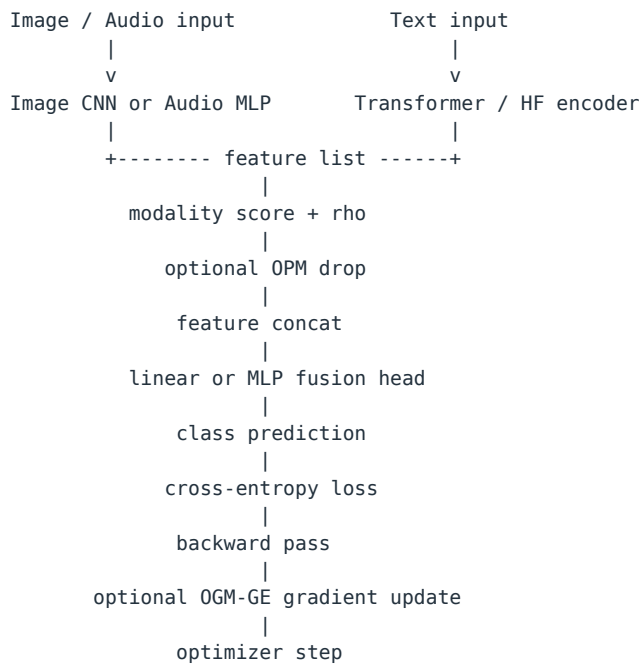
If text is easier, its branch can reduce the classification loss quickly. Once the fused prediction is already confident, the loss gradient becomes small. The image branch then receives a weak learning signal and may remain under-optimized. This is the **rich-get-richer effect** behind modality imbalance.

The project follows this cycle:

Detect -> Quantify -> Mitigate -> Verify

- **Detect:** attention, Integrated Gradients, gradient $\times$ input, and uni-modal probes
 - **Quantify:** discrepancy ratio ρ
 - **Mitigate:** OPM and/or OGM-GE
 - **Verify:** fused accuracy, uni-modal accuracy, ρ , macro-F1, ECE, and plots
-

3. High-level architecture



Encoders

- **Image:** [SmallCNN](#) or ResNet-18
- **Text:** local Transformer encoder or optional Hugging Face model
- **Audio:** [VectorMLPEncoder](#) for precomputed feature vectors
- **Fusion:** concatenation followed by a linear or MLP classification head

The data loader automatically detects the modalities from CSV columns. For example, a CSV with [audio](#), [text](#), [label](#) builds an audio/text model without an image branch.

4. Core measurement: discrepancy ratio

For each modality m , the model first obtains that modality's contribution to the classifier logits. For a linear head:

$$\text{unimodal_logit}_m = W_m * \phi_m + \text{bias} / M$$

The true-class confidence of modality m for sample i is:

$$s_i^m = \text{softmax}(\text{unimodal_logit}_m)[\text{true_class}_i]$$

The batch discrepancy ratio is:

$$\rho^m = \text{average over } j \neq m \text{ of:} \\ \sum_i(s_i^m) / \sum_i(s_i^j)$$

Interpretation:

- $\rho = 1$: modality scores are balanced
- $\rho > 1$: this modality is dominant
- $\rho < 1$: this modality is weaker

[bml/modulation.py](#) implements the score and ratio calculations. [bml/engine.py](#) records them during both training and evaluation.

5. Mitigation methods

5.1 OPM: On-the-fly Prediction Modulation

OPM acts in the **forward pass**. It calculates an adaptive feature-drop probability for a dominant modality:

$$\begin{aligned} z &= \tanh(\rho - 1) \\ q &= q_{\text{base}} * (1 + \lambda * z), \text{ if } \rho > 1 \\ q &= 0, \text{ otherwise} \end{aligned}$$

A dominant feature is randomly removed for selected training samples. The weak modality is not dropped. This prevents the dominant branch from solving every sample by itself and allows the weak branch to learn.

Important implementation detail: the current q is applied first, and the latest ρ updates q for the next iteration.

5.2 OGM: On-the-fly Gradient Modulation

OGM acts in the **backward pass**. It scales the dominant encoder's gradient:

$$\begin{aligned} k &= 1 - \alpha * \tanh(\rho - 1), \text{ if } \rho > 1 \\ k &= 1, \text{ otherwise} \end{aligned}$$

- dominant encoder: gradient is reduced by factor k
- weak encoder: gradient is unchanged
- fusion head: not modulated

This slows the dominant branch rather than artificially increasing the weak branch's gradient.

5.3 GE: Generalization Enhancement

OGM-GE also adds Gaussian noise based on each parameter tensor's gradient standard deviation. The purpose is to recover useful stochasticity and improve generalization after gradient scaling.

5.4 Combined mode

--modulation both applies OPM in the forward pass and OGM-GE after backpropagation.

Combined mode is strongest on the synthetic benchmark, but it is not guaranteed to be best on every dataset.

6. Exact training flow

For every mini-batch, `train_one_epoch()` performs these steps:

- 1** Move inputs and labels to the selected device.
- 2** Run each modality encoder once.
- 3** Compute uni-modal logits from the undropped features.
- 4** Compute true-class scores and `rho`.
- 5** Apply OPM feature dropping when enabled.
- 6** Concatenate features and calculate fused logits.
- 7** Compute cross-entropy loss.
- 8** Run backpropagation.
- 9** Apply OGM gradient scaling and GE noise when enabled.
- 10** Apply optional gradient clipping.
- 11** Run the optimizer step.
- 12** Log loss, accuracy, uni-modal accuracy, `rho`, `q`, and `k`.

The best validation checkpoint and final checkpoint are stored in the run directory. Model checkpoints are Git-ignored, while configuration, history, summary, vocabulary, label map, and logs are retained for reproducibility.

7. Repository map

Path	Responsibility
Project_Proposal.md	original problem statement, objectives, and proposed BMA extension
multimodal (2).ipynb	exploratory PaliGemma inference and attention heat-map notebook
balanced_mm/train.py	CLI arguments, model construction, optimizer, scheduler, and training entry point
balanced_mm/eval.py	checkpoint evaluation, confusion matrix, macro-F1, ECE, and probes
balanced_mm/compare.py	side-by-side comparison of completed runs
balanced_mm/plot_history.py	training and discrepancy curves
balanced_mm/bml/data.py	tokenization, CSV loading, audio features, synthetic dataset, and data loaders
balanced_mm/bml/models.py	image, text, audio encoders and late-fusion model
balanced_mm/bml/modulation.py	discrepancy ratio, OPM, OGM-GE, and unified controller
balanced_mm/bml/engine.py	training, evaluation, and frozen-encoder probing
balanced_mm/bml/xai.py	Integrated Gradients, token relevance, and modality attribution
balanced_mm/bml/metrics.py	accuracy, confusion matrix, per-class accuracy, macro-F1, and ECE
balanced_mm/bml/analysis.py	aggregates runs and generates report tables/plots
balanced_mm/scripts/	dataset preparation, benchmark sweeps, report generation, notebook build
balanced_mm/tests/	tests for equations, models, data, XAI, metrics, and report generation
balanced_mm/runs/	raw run configuration, history, summary, and logs
balanced_mm/results/	aggregate RESULTS.md and benchmark figures

8. Datasets and experimental setup

8.1 Synthetic benchmark

- modalities: generated image + generated text
- classes: 6
- training / validation: 2,400 / 600
- informative probability: 0.7 independently for each modality
- image signal: class-specific colored patch over Gaussian noise
- text signal: class-specific keyword among random tokens
- model: SmallCNN + two-layer Transformer
- training: 15 epochs, SGD, learning rate 0.01, batch size 32
- reporting: three seeds

The theoretical uni-modal ceiling is approximately 0.75, while the fused ceiling is approximately 0.925. A high fused score therefore requires learning both branches.

8.2 Food-101 experiment

- modalities: image + constructed prompt text
- subset: 10 classes
- training / validation: 2,500 / 1,000
- image encoder: SmallCNN at 64 × 64 model input
- text construction: correct class prompt with probability 0.7; generic food prompt otherwise
- training: 12 epochs, three seeds

Food-101 has no native text annotation. The text modality is deliberately constructed for a controlled imbalance experiment and must not be described as an original Food-101 annotation.

8.3 MELD experiment

- modalities: dialogue text + official 300-dimensional audio features
- classes: 7 emotions
- official split sizes: 9,989 train / 1,109 validation / 2,610 test
- text encoder: local Transformer, maximum length 48
- audio encoder: MLP over precomputed features
- training: 12 epochs, three seeds

The experiment uses precomputed audio features, not end-to-end waveform learning.

9. Main results

All values below are mean $\pm$ standard deviation over three seeds. ρ values are final training discrepancy ratios from the generated report.

Synthetic

Method	Best validation accuracy	Final ρ_{image}
No modulation	0.887 $\pm$ 0.022	1.344
OGM-GE	0.898 $\pm$ 0.017	1.188
OPM	0.904 $\pm$ 0.023	1.140
OPM + OGM-GE	0.906 $\pm$ 0.025	1.121

Meaning: all modulation methods reduce image dominance and improve validation accuracy. Combined modulation gives the best mean accuracy and lowest image discrepancy in this experiment.

Food-101

Method	Best validation accuracy	Final ρ_{text}	Final ρ_{image}
No modulation	0.812 $\pm$ 0.004	2.726	0.381
OGM-GE	0.814 $\pm$ 0.005	2.694	0.385
OPM	0.845 $\pm$ 0.000	2.085	0.494
OPM + OGM-GE	0.842 $\pm$ 0.005	2.005	0.512

Meaning: prompt text dominates because it frequently contains the class name. OPM gives the best accuracy improvement, while combined modulation gives the smallest modality gap.

MELD

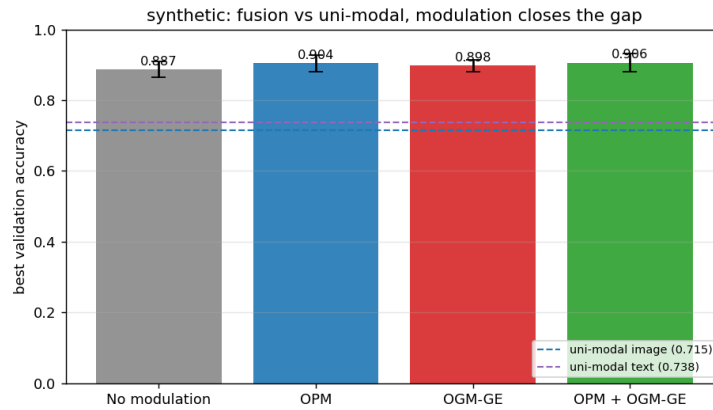
Method	Best validation accuracy	Final ρ_{text}	Final ρ_{audio}
No modulation	0.578 $\pm$ 0.002	2.219	0.458
OGM-GE	0.573 $\pm$ 0.002	1.598	0.632
OPM	0.565 $\pm$ 0.004	1.237	0.819
OPM + OGM-GE	0.561 $\pm$ 0.006	1.219	0.832

Meaning: modulation strongly balances text and audio, but validation accuracy decreases. Text alone is already strong (0.560) and audio is weaker (0.469), so aggressively limiting text can reduce overall predictive performance. This demonstrates an accuracy–balance trade-off.

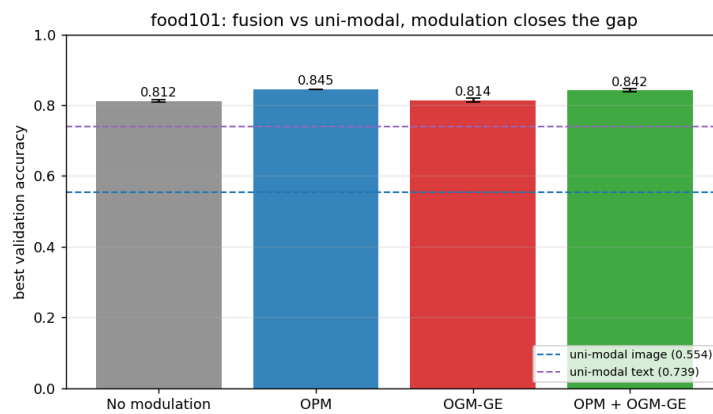
Benchmark figures

Bars best validation accuracy dikhate hain. Exact mean $\pm$ standard deviation values Section 9 aur `balanced_mm/results/RESULTS.md` me available hain.

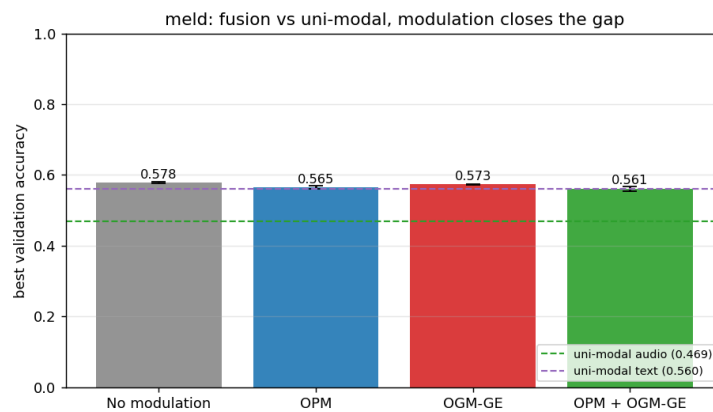
Synthetic benchmark



Food-101



MELD



10. XAI and verification tools

PaliGemma notebook

`multimodal (2).ipynb` is an exploratory notebook that loads PaliGemma, accepts an image and prompt, requests generation-time attention, and overlays image-token attention as a heat map. It demonstrates the detection side of the proposal; it does not apply OPM/OGM end-to-end to the 3B model.

Integrated Gradients

`integrated_gradients_image()` interpolates from a zero-image baseline to the input image and accumulates predicted-class gradients. The result has the same shape as the image and estimates pixel contribution.

Text relevance

`token_importance_text()` uses gradient $\times$ embedding because gradients with respect to discrete token IDs are not meaningful. Padding positions are masked out.

Modality attribution

`modality_attribution()` converts absolute image and text attribution into normalized shares that sum to one. These shares provide an XAI view of relative modality contribution.

Uni-modal probing

`--probe` freezes each trained encoder, extracts its features, and fits a new linear classifier. Probe accuracy measures encoder quality independently of the existing fusion head.

11. How to run the project

Install

```
cd balanced_mm
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
pip install pytest
```

Single synthetic run

```
python train.py \
  --data synthetic \
  --image_encoder smallcnn \
  --img_size 64 \
  --epochs 15 \
  --syn_train_n 2400 \
  --lr 0.01 \
  --modulation ogm \
  --out_dir runs/syn_ogm
```

Complete synthetic comparison

```
bash scripts/run_synthetic_compare.sh 15 2400 smallcnn 3
bash scripts/run_seed_sweep.sh
python scripts/make_report.py
```

Food-101

```
python scripts/make_food101_csv.py --root data/food101
bash scripts/run_food101.sh
python scripts/make_report.py
```

MELD

```
python scripts/make_meld_csv.py \
  --features_dir <meld-features-directory> \
  --out data/meld
bash scripts/run_meld.sh
python scripts/make_report.py
```

Evaluate and compare

```
python eval.py --run_dir runs/ogm --test_csv data/test.csv --probe
python compare.py runs/none runs/opm runs/ogm runs/both
python plot_history.py runs/none runs/opm runs/ogm runs/both --out curves.png
python -m pytest tests
```

12. Output files

Each training run can contain:

File	Meaning
config.json	exact CLI configuration
history.json	epoch-wise training and validation metrics
summary.json	best accuracy, best epoch, and final metrics
train.log	readable training log
vocab.json	fitted word-level vocabulary
label_map.json	class-name to integer mapping

File	Meaning
<code>best.pt</code>	best validation checkpoint; Git-ignored
<code>last.pt</code>	final checkpoint; Git-ignored

`python scripts/make_report.py` scans completed runs and generates `results/RESULTS.md` plus accuracy and discrepancy plots.

13. What is implemented vs proposed

Implemented

- discrepancy monitoring for two or more modalities
- OPM, OGM, and GE
- image/text and audio/text late fusion
- synthetic, Food-101, and MELD pipelines
- XAI attribution utilities
- PaliGemma attention demonstration
- metrics, tests, seed sweeps, run logs, and automatic reports

Proposed or future work

- Balanced Modal Attention (BMA) inside PaliGemma cross-attention
- XAI-driven regularization directly on PaliGemma
- full VQAv2 / COCO evaluation
- end-to-end modulation of the 3B PaliGemma model
- full-resolution and larger-backbone experiments
- end-to-end MELD waveform training

This distinction is important: OPM/OGM-GE are the implemented mitigation baselines; BMA remains part of the proposal.

14. Limitations

- 1 Food-101 text is constructed rather than naturally annotated.
- 2 Food-101 uses a compact ten-class, low-resolution protocol.
- 3 MELD audio is represented by precomputed features.
- 4 Lower discrepancy does not guarantee higher task accuracy.
- 5 Hyperparameters are not exhaustively tuned for every dataset.
- 6 PaliGemma is used for exploratory detection, not full mitigation training.
- 7 Attention is useful for visualization but should not be treated as a complete causal explanation.

15. Short explanation script

This project studies modality bias in multimodal learning. In joint training, one modality can become dominant and suppress learning in the weaker encoder. The repository detects this using attention, Integrated Gradients, uni-modal scores, and a discrepancy ratio called ρ . It mitigates the problem using OPM, which adaptively drops dominant features in the forward pass, and OGM-GE, which scales dominant encoder gradients and adds controlled noise in the backward pass. The methods are evaluated on synthetic image/text data, a controlled Food-101 image/prompt setup, and MELD text/audio features. Modulation improves both balance and accuracy on synthetic and Food-101 data. On MELD it improves balance but slightly reduces accuracy, showing that balance and task performance can trade off when one modality is genuinely more informative.

16. Key terms

Term	Meaning
Modality	one type of input, such as image, text, or audio
Modality bias	excessive dependence on one modality
Late fusion	concatenate modality features before classification
Uni-modal logit	prediction contribution from one modality
ρ	relative true-class confidence of one modality vs the others
OPM	adaptive dominant-feature dropping in the forward pass
OGM	adaptive dominant-gradient scaling in the backward pass
GE	gradient-noise regularization used with OGM
Integrated Gradients	attribution from a baseline to the actual input
ECE	Expected Calibration Error
Probe	new classifier trained on frozen encoder features

Reference

Y. Wei, D. Hu, H. Du, and J.-R. Wen, "On-the-fly Modulation for Balanced Multimodal Learning," IEEE Transactions on Pattern Analysis and Machine Intelligence, 2024.